Multi-Head Self Attention

Combining several self attention mechanisms give the self attention greater power of discrimination

Each Head is indexed with "**r**"

Each head has its own matrices $\mathbf{W}^r_q$, $\mathbf{W}^r_k$, $\mathbf{W}^r_v$.

For input **X** each attention head produces a different output vector $\mathbf{Y}^r$.

All Output vectors are concatenated , and pass through a **linear transformation** to reduce the dimension back to **original X** dimension.